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Using Signal Decomposition Methods and Deep Learning Approaches to Forecast Bitcoin Price

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Adaptive Noise (CEEMDAN), Variational Mode Decomposition (VMD), and Ensemble Patch Transform (EPT)—along with two composite signal decomposition methods—EPT-CEEMDAN and EPT-VMD—to decompose the original hourly Bitcoin price data from 2019 to 2023 into several sub-components, which are then individually input into deep learning models using Long Short-Term Memory (LSTM), Bidirectional Long Short-Term Memory (BiLSTM) and Temporal Convolutional Networks (TCN) for training and prediction. Finally, the predicted values of all sub-components are reconstructed to produce a final price prediction, and to compare the prediction error rates of various experimental combinations. Experimental results show that in most cases, combining signal decomposition methods can effectively improve model prediction performance, and the EPT-VMD-TCN method proposed in this research produced the best comprehensive prediction performance, with MAPE error rates of 0.0014, 0.004, 0.0007, 0.0076 and 0.0011 respectively for the 2019–2023 years.

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Appendices

Appendix A. Variational Mode Decomposition (VMD)

The VMD algorithm assumes that all signals can be composed of a set of limited-bandwidth mode functions (modes) with a central frequency, similar to the IMF in EMD. The core of the VMD algorithm seeks to solve a variation problem by decomposing the original signal into K modal functions, and requires the sum of the bandwidths of the center frequencies of each

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To convert the constrained variation problem of the above equation into an unconstrained variation problem, the quadratic penalty factor α and the Lagrangian operator λ are introduced, and the above formula is expanded as shown in Eq. (8):

$$\begin{aligned} \mathcal{L}(u_k, \lambda) = & \alpha \sum_k \left| \frac{\partial}{\partial t} \left[\left(\delta \left(t \right) + \frac{j}{\pi t} \right) * u_k \left(t \right) \right] e^{j \omega_k t} \right|^2 \\ & + \left| f \left(t \right) - \sum_k u_k \left(t \right) \right|^2 + \left\langle \lambda \left(t \right), f \left(t \right) - \sum_k u_k \left(t \right) \right\rangle \end{aligned} \quad (8)$$

Then the Alternating Direction Method of Multipliers (ADMM) is used to iteratively update

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Appendix B. Ensemble Patch Transform (EPT)

The EPT algorithm assumes that in the patch process, relevant data blocks for each time point t in the time series data are given, and the local structure of the data is identified based on block size. The EPT algorithm provides two types of blocks: rectangular and oval. Oval blocks are well suited for capturing the central tendency of the data, but rectangular blocks are better suited to capturing sudden changes in the data in highly unstable signals. Given the data sampling period τ , in the time series data $\{X_t\}$, the time point t is the center point of the block $\{(t, X_t)\}$ located from point $(\left(t+k, \underset{\text{\textit{k}}}{\epsilon} \left[-\tau/2, \tau/2\right]\right)\{\text{min}\} \{X_{t+k}-0.5\gamma \tau \text{right})\}$ to point $(\left(t+k, \underset{\text{\textit{k}}}{\epsilon} \left[-\tau/2, \tau/2\right]\right)\{\text{max}\} \{X_{t+k}+0.5\gamma \tau \text{right})\})$, forming a closed rectangle with a width equal to τ , and the height of the block can be expressed as Eq. (12):

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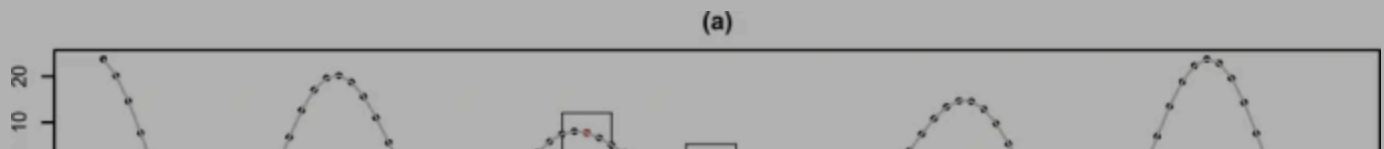
$$\mathbb{U}_t^{\tau}(\mathbf{X}_t) = \underset{\mathbf{k} \in [-\tau/2, \tau/2]}{\mathop{\mathrm{min}}}\{\mathbf{X}_{t+k} + 0.5\gamma\tau\}$$

(14)

$$\mathbb{M}_t^{\tau}(\mathbf{X}_t) = \frac{1}{2}(\mathbb{L}_t^{\tau}(\mathbf{X}_t) + \mathbb{U}_t^{\tau}(\mathbf{X}_t))$$

(15)

Fig. 19



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In the ensemble process, in which blocks are moved within a fixed period τ to produce ensemble blocks that effectively represent the temporal changes in the data by enhancing their temporal resolution. The ensemble block transformation is defined as follows: for a fixed period τ , the (l_{th}) offset block at time point t is defined as $(P_{t+\ell}^{\tau}(X_t), \ell \in [-\tau/2, \tau/2])$, then the time set of all possible offset blocks for point t can be defined as $(EP_t^{\tau}(X_t) := \{P_{t+\ell}^{\tau}(X_t), \ell \in [-\tau/2, \tau/2]\})$. Finally, by compiling relevant statistical data, we can obtain the high-frequency mode $(HF_t^{\tau}(X_t))$ and the low-frequency mode $(EM_t^{\tau}(X_t))$, as shown in the Eqs. (16–17).

$$HF_t^{\tau}(X_t) = X_t - EM_t^{\tau}(X_t) \quad (16)$$

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$$f_i = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

(19)

$$f_o = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

(20)

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

(21)

Appendix D. Temporal Convolutional Networks (TCN)

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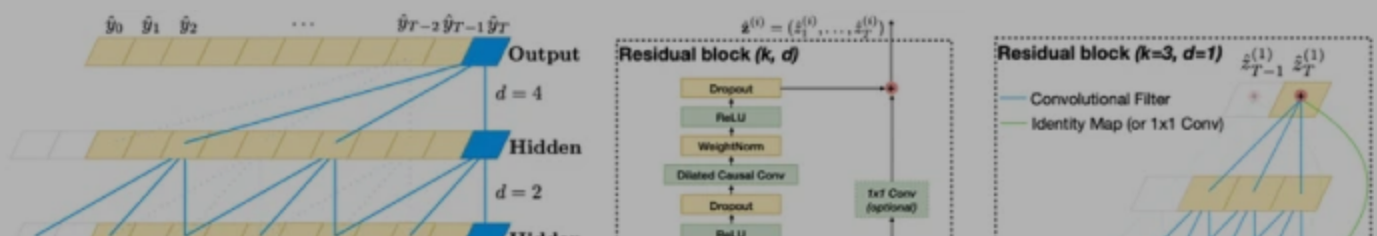
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The TCN receptive field size depends on the network depth, the filter size k and the dilation factors d , where each layer contains multiple filters for extracting features. Networks with greater numbers of layers require greater stability, so TCN replaces the convolution layer with the residual block, as shown in Fig. 20b. In addition, the number of channels of the input and output values in TCN may differ. For this reason, TCN adds a 1×1 convolution to ensure that the element-level addition $\oplus$ has the same number of channels, as shown in Fig. 20c.

Fig. 20



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